RWTH Aachen University · M.Sc. Robotic Systems

Internship Report

Vision Guard: A Vision-Based PPE Compliance Interlock for Flat Belt Grinding

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Confidential: prepared for academic assessment only. Part numbers, part geometry, production volumes, cycle times, supplier names, commercial terms, network addresses, hostnames, credentials and other employee names are excluded throughout. No image shows a manufactured component or an engineering drawing: the station layouts are simplified models, and the only person shown is the author, who consented.

Acknowledgements

I am grateful to my industrial supervisor, [name, designation], for entrusting me with a project that began as an observation on the shopfloor rather than a prepared brief, and for the freedom to carry it from the first prototype to a plant-wide rollout decision. I thank [name, designation of subject-matter expert(s)] for the time they gave to explaining the grinding operation, and for their candour about what the shopfloor would and would not accept.

The Environment, Health and Safety team reviewed the system through the Management of Change process and helped me see it as a control within a wider safety system rather than as a stand-alone device; their questions led to the failure analysis in Section 6.4. The maintenance department carried out the electrical commissioning, the installation and the later relocation of the station, and their practical knowledge of the machines helped me avoid several errors. The IT department helped me fit the industrial PC into the site’s standard configuration.

I owe a particular debt to the operators of the flat belt grinders. They tolerated a camera system watching them work, told me plainly when it stopped them for no reason, and showed me how the operation is actually performed. The most consequential finding of this project (Section 8.3) came from them.

Abstract

During my internship at GE Aerospace in Pune, India, I contributed to the design, development and commissioning of **Vision Guard**, a camera-based interlock that turns PPE compliance at a flat belt grinding station from an administrative control into an engineering control. Two cameras observe the operator. An industrial PC without a discrete GPU runs a YOLO11s detector compiled with OpenVINO and decides whether the required PPE is worn. An Ethernet relay module, addressed over Modbus TCP, drives a tower light and buzzer and grants or withholds the grinder motor's run permissive.

Migrating inference from PyTorch to OpenVINO cut end-to-end latency from 300–400 ms to about 40 ms. Six model versions were trained. The first deployed version reached a validation precision of 0.942, a recall of 0.955 and an mAP@0.5 of 0.966, and the station now runs an eight-class version that adds the apron. On a common test set of 477 station images, that version was the most accurate of the six (mAP@0.5 0.948). The system passed two Management of Change approvals, was installed and relocated, and was approved for deployment on every flat belt grinder at the plant.

Observing the system in use showed that operators stand at an angle to the machine to inspect their work, outside the geometry the cameras were placed for, and this changed how camera placement is specified for the rollout. A failure-mode review found two faults in the software's fail-safe behaviour, which I fixed, and confirmed that the emergency stop needs a hard-wired safety relay to act independently of the software.

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1. Introduction

1.1 Motivation

Flat belt grinding is a manual finishing operation. The operator holds a component by hand against a moving abrasive belt. The hazards are three: abrasive dust, sparks, and contact injury to the hands and forearms.

The station's required PPE answers each of them: safety glasses, gloves, sleeves and a mask. The deployed system also checks a head net, which the site requires at the station.

Before this project, compliance was assured by training, signage and supervision. In the hierarchy of controls [1] these are all *administrative* controls. Each depends on a person's decision at the moment of use, and on a supervisor being present. None of them acts on the machine. When the decision goes the wrong way, the grinder runs regardless. Vision Guard moves the control one tier up: it detects missing PPE from camera images and inhibits the machine. It does not replace the machine's guarding or its emergency stop. It closes the gap between the rule that PPE must be worn and the fact of the machine running.

1.2 Host organisation

GE Aerospace designs and manufactures commercial and military aircraft engines, components and systems. It has operated as an independent, publicly listed company since April 2024.

The GE Aerospace Multi-modal Manufacturing Facility (MMF) in Pune is a global centre of excellence for high-precision, mission-critical structural components, such as ducts and manifolds, for commercial jet engines including the LEAP, GEnx and GE9X. Its processes include five-axis CNC machining, specialised welding, hydroforming and a digitally connected production infrastructure. I completed the internship in the Manufacturing Engineering section of the Fabrication Center of Excellence (FCOE) department, which programs, optimises and deploys robotic and automated cells on the shopfloor.

1.3 Origin, scope and objective

The project did not exist when the internship began. In my first two weeks I studied the shopfloor systematically, covering:

- material flow and the process sequence;
- weld preparation and manual welding;

- the robotic TIG welding cell.

Throughout, I spoke with welders, operators and engineers. The grinding station emerged from that study as a candidate for an engineering control. The project came from observation, not from a prepared brief.

The objective was a system that **withholds the grinder's run permissive unless the operator at the machine wears the required PPE**. It had to do so quickly enough, and steadily enough, to be accepted by the people who work at the machine. The work covered the whole life of the system:

- dataset, model and inference pipeline;
- compliance logic and interlock;
- hardware, commissioning and safety approval;
- installation, relocation and operator documentation;
- the specification of a plant-wide rollout.

Chapter 9 describes the plant-wide rollout, and Chapter 10 reflects critically on the internship.

2. Technical background

2.1 Compliance is a decision, not a detection

An object detector answers *is there a glove in this image, and where?* An interlock must answer *is this operator, now, protected as the site rules require?* The difference between the two questions accounts for most of the engineering in this report. It has three dimensions.

- **Ownership.** A glove on a bystander or on a bench protects nobody. The decision needs to know *whose* equipment is judged.
- **State.** An object can be present and still not compliant. A glove can be held in the hand instead of worn. Prescription spectacles can be worn instead of safety glasses. A sleeve can be worn so that it can be caught. In each case the detector may be entirely right about the object, and the compliance decision still wrong.
- **Time.** A glove hidden for a few hundred milliseconds behind the workpiece has not been removed. A single compliant frame does not prove that the operator is compliant now. The decision must integrate evidence over time.

Validation metrics measure only the detection question. They are necessary but not sufficient.

2.2 Detector and runtime

YOLO [2] is a single-stage detector: one network pass predicts boxes and class scores for the whole image, so the inference cost is fixed and predictable. YOLO11 [3] is the Ultralytics release used here, in its *s* scale (about 9.4 million parameters). Detection transformers such as DETR [4] and RF-DETR [5] predict object sets directly and need no non-maximum suppression [6], but their attention layers are costly on a CPU.

OpenVINO [7] compiles a trained network for a specific CPU. Detection quality is reported as precision , recall and mean average precision (mAP) at an IoU threshold of 0.5, and averaged over thresholds from 0.5 to 0.95 [8], [9], [10].

2.3 Two error directions

An interlock can be wrong in two directions, which Table 1 sets out. A single accuracy figure would weight them equally, which is wrong for both.

Table 1. Error directions of the PPE interlock.

Decision truth	Operator protected	Operator not protected
Machine permitted	Correct	**False permit**: hazard present while the machine runs
Machine inhibited	**False stop**: compliant operator stopped	Correct

A false stop usually follows a *missed detection* (a false negative) of equipment that is worn. Its cost is lost production, and with it the operator's trust: a station that stops compliant people gets worked around. A false permit is the exposure the system exists to prevent. Every parameter in Chapter 6 is a position on this trade-off.

Such a system is a **permissive**: a condition that must hold before the machine may run, layered on top of the machine's safety functions and never in place of them [11]. It talks to the relay module over Modbus TCP [12], with the `pymodbus` library [13]. Modbus is stateless and driven by the client. A device keeps its outputs as last written until it is told otherwise, unless it has its own communication watchdog.

3. System overview

Figure 1 shows the system as built. Two USB cameras view the operator from the front and the side. The industrial PC, which runs on its CPU only, carries the whole software stack:

- capture;
- detection;

- the compliance logic;
- the motor-permissive latch;
- an operator interface on a monitor at the station.

The relay module's outputs drive the green, amber and red lamps, the buzzer and the motor permissive (DO1–DO5). Its inputs read the emergency stop and a start/stop push button (DI1, DI2). A control panel houses the relay module, its power supply, the e-stop, the push button and the PC. An overload relay protects the motor, and the camera stands were fabricated to measured dimensions. Table 2 gives the module's Modbus map.

Table 2. Modbus map of the relay module (addresses 0-based).

Signal	Object	Address	Channel	1 means
Green / amber / red lamp	Coil	0 / 1 / 2	DO1–DO3	lamp on
Buzzer	Coil	3	DO4	sounding
Motor permissive	Coil	4	DO5	permissive granted
E-stop	Discrete input	0	DI1	operated (normally open)
Push button	Discrete input	1	DI2	pressed (normally open)

Lamps: red means an e-stop or a violation; green means compliant and running; amber means compliant and idle, or DEGRADED; dark means STANDBY.

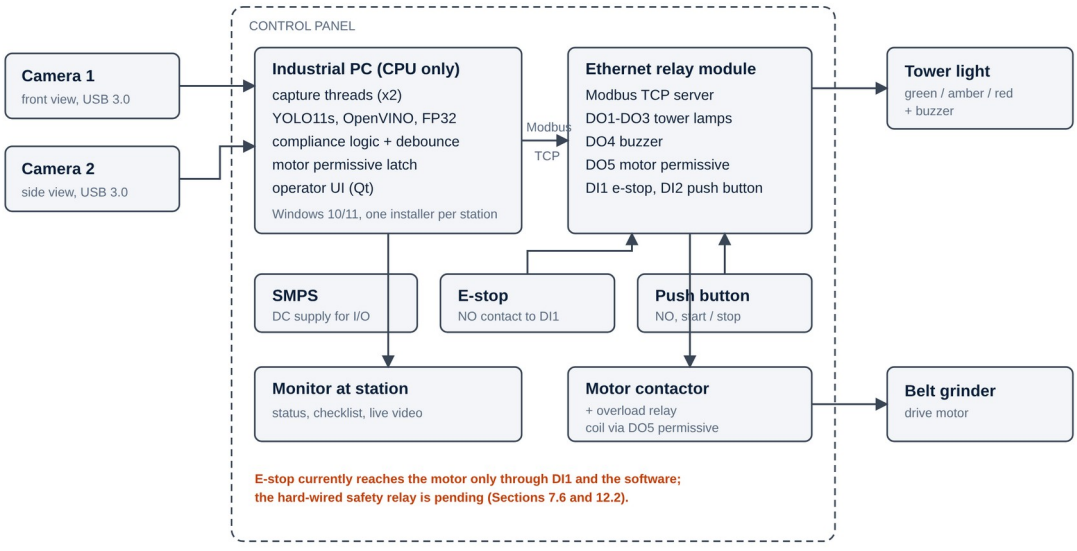


Figure 1. System architecture.

The software is a Python application of about 5,800 lines, with a test suite, held in the project repository (`ppe/`, `main.py`, `config.yaml`). Its main dependencies are:

- Ultralytics;

- OpenVINO;
- OpenCV;
- PySide6 (Qt);
- pymodbus.

Library versions are pinned for a Windows installer, so that every station runs an identical build. Each cycle takes one camera frame through letterboxing, inference, the compliance decision and the relay write. The result is shown over the live video, with a checklist of the required items and a countdown whenever a correction window is running.

The emergency stop is wired as a normally open contact into DI1. Until the planned safety relay is fitted, this input is the only thing the e-stop does, so it acts through the software. Section 6.4 returns to this point.

4. Dataset and model

4.1 Class schema and site-specific data

The model described in this report detects seven classes (Table 3). They are five items of equipment, the person who wears them, and one *non-compliant state*, `Wrong Sleeve`, which the station treats as a violation whenever it is detected. The station now runs a later version, `best-apron`, that adds an `Apron` class (Section 4.4). Section 4.3 describes the training of the seven-class model that was deployed first.

Table 3. Class schema and the role of each class.

Class	Role in the decision	Count required
Gloves	must be worn	1 (2 before station tuning)
sleeves	must be worn, or an apron instead	1 (2 before station tuning)
Safetyglasses, Mask, headnet	must be worn	1 each
Wrong Sleeve	forbidden: detecting it is the violation	0
person	identifies whom to judge	—
Apron	the sleeves alternative; in the retrained station model	1

All training images were either captured on the shopfloor or taken of working poses recreated in an isolated environment, and no public PPE dataset was used. Public datasets centre on construction-site equipment [14], [15], [16]. The station's lighting, background, operator postures and issued PPE are particular to it, and sleeves are absent from most public sets. Starting from COCO-pretrained weights and fine-tuning on site data gave the model general features from a large dataset, and everything station-specific from the station. Images were

deliberately taken from varying camera positions, which proved valuable when the installation was moved (Section 7.2).

4.2 Local annotation and failure-driven expansion

Company data compliance rules forbid uploading shopfloor images to cloud annotation services. I therefore built an **auto-annotation tool** that pre-labels images locally with the current model. An annotator corrects the boxes, following written annotation guidelines, rather than drawing them all, which is several times faster. One remote day yielded 200 annotated images. The station software later gained a **capture mode** that keeps frames worth training on during normal operation, pre-labelled, and never sends them off the site network. Its triggers are:

- a timer;
- a violation;
- a motor start or stop;
- a required item going out of sight.

The dataset grew in rounds to about 5,000 images, each round aimed at failures found in systematic testing of the previous model. The failures fall into two categories, and the distinction is the conceptual core of the data work.

- **Detection failures** are cases where the model misses or confuses an object that is there. Examples are partly occluded PPE, unusual poses, confusion between gloves and sleeves at the wrist, missed items in the side view after the relocation, and erratic detection of people in the background. More annotated examples of the hard case are the right fix.
- **Decision failures** are cases where the object is detected *correctly* and the compliance decision is still wrong. Examples are prescription spectacles worn in place of safety glasses, and gloves carried rather than worn. More data with the same labels cannot fix them, because the labels do not encode the distinction. There are only two fixes. The schema can label the non-compliant state explicitly, or the decision can test whether the item lies on the body part it should cover.

Leaving the non-compliant case unlabelled teaches the model weakly to ignore it. More seriously, it makes the dangerous error unmeasurable, because a missed worn glove and a correctly ignored carried glove both produce *no box*.

The schema takes the first route for sleeves: `Wrong Sleeve` is an explicit class with the strictest settings in the system. For glasses and gloves, neither route is in place yet. The relational test checks only that an item's box lies inside the person's box, relaxed to 10 % containment for gloves, and a glove held in the hand passes it.

4.3 Training and convergence

The model is YOLO11s, fine-tuned from COCO weights:

- input 640 × 640, batch size 16;
- cosine learning-rate schedule from 0.01 [17];
- mosaic [18], colour and ±5° rotation augmentation;
- up to 150 epochs, with early stopping after 30 epochs without improvement.

I trained it on a CUDA GPU in about two hours. Table 4 lists the full configuration.

Table 4. Training configuration of the first deployed model (v2).

Parameter	Value	Parameter	Value
Initial weights	`yolo11s.pt` (COCO)	Image size	640 (square, `rect: false`)
Epochs planned / run	150 / 105	Patience	30 (best epoch 75)
Batch	16	Optimiser	`auto`
`lr0` / `lrf`	0.01 / 0.01, cosine	Momentum / weight decay	0.937 / 0.0005
Warm-up	3 epochs	Loss gains box / cls / dfl	7.5 / 0.5 / 1.5
HSV h / s / v	0.015 / 0.7 / 0.4	Rotation / translate / scale	±5° / 0.1 / 0.5
Flip left–right / up–down	0.5 / 0	Mosaic / mixup / copy-paste	1.0 / 0 / 0
Auto-augment / erasing	RandAugment / 0.4	AMP / seed	on / 0 (deterministic)
Device	CUDA GPU 0	Duration	7,180 s (≈ 2 h)
Ultralytics (train / export / run)	8.4.126 / 8.4.130 / 8.3.253	Export	OpenVINO IR, FP32, batch 1, static 640 × 640

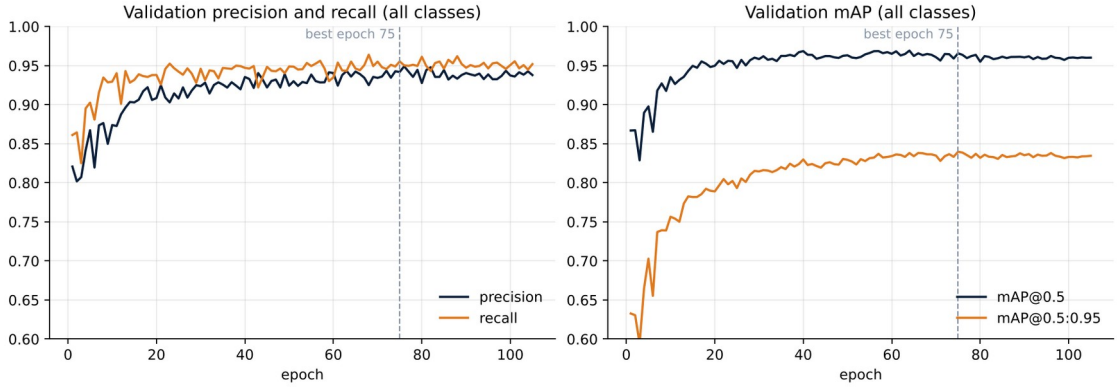


Figure 2. Validation precision, recall and mAP over 105 epochs.

Figure 2 shows the convergence in three phases:

1. **Rapid adaptation.** The pretrained network reached an mAP@0.5 of 0.867 after one epoch.
2. **Steady gains in localisation.** mAP@0.5 reached 0.96 by about epoch 40, and mAP@0.5:0.95 kept rising until epoch 75.
3. **Plateau.** The validation losses stopped improving while the training losses kept falling (Figure 3), the characteristic onset of over-fitting.

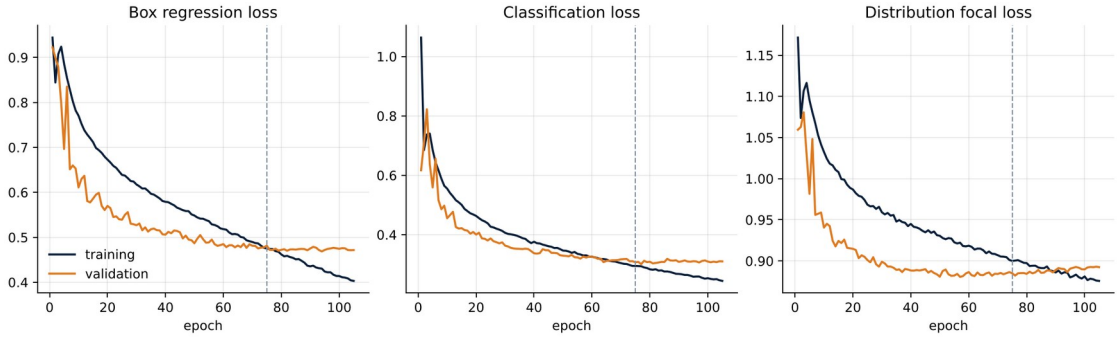


Figure 3. Training and validation losses of v2; dashed line at the best epoch.

Early stopping ended training at epoch 105 and kept epoch 75. At that epoch the model reached precision 0.942, recall 0.955 (F1 0.949), mAP@0.5 0.966 and mAP@0.5:0.95 0.840, averaged over the seven classes.

Mosaic switch-off and the final low-learning-rate phase were scheduled against the *planned* 150 epochs, so early stopping meant they never ran. Both can be recovered at no cost at the next retraining.

4.4 Model selection

Table 5 lists the configurations evaluated.

Table 5. Configurations evaluated on the station hardware.

Model	Runtime	End-to-end latency	Outcome
YOLO11s	PyTorch	300–400 ms (inference 200–300 ms)	superseded
YOLO11s	OpenVINO	≈ 40 ms, within budget	**deployed**
RF-DETR	—	≈ 1500 ms	rejected on latency; accuracy good

Model versions. I trained six versions in all (Table 6), one of them on the smaller YOLO11n scale. Figure 4 shows their training curves.

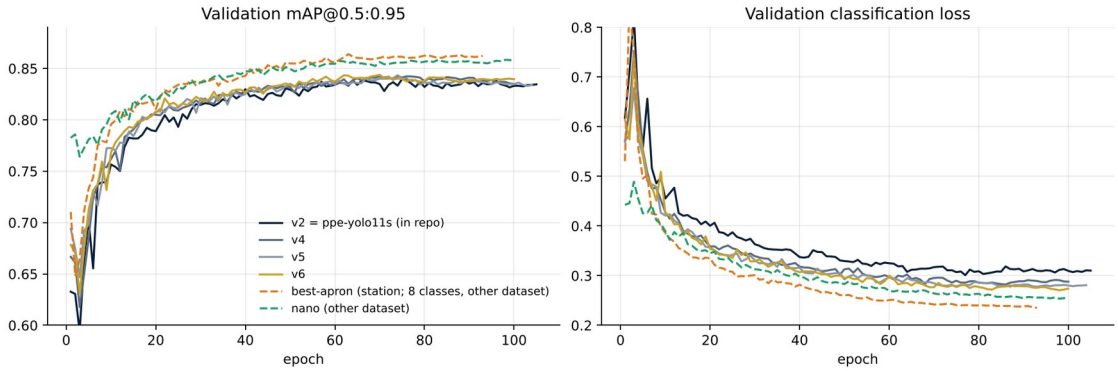


Figure 4. Validation mAP@0.5:0.95 and classification loss of the six model versions. Dashed lines were trained on different data and are not directly comparable.

Table 6. Model versions: validation metrics at each run's best epoch, and CPU inference time after conversion to OpenVINO with the same tool (FP32, 640 × 640, median of 60 calls on one 4-core CPU).

Version	Trained	Classes	Epochs (best)	P	R	mAP@0.5	mAP@0.5:0.95	OpenVINO ms
v2 (first deployed)	23 Aug	7	105 (75)	0.942	0.955	0.966	0.840	35.0
nano 3	30 Aug	7	100 (98)	0.969	0.967	0.985	0.858	16.5
v4	10 Sep	7	100 (70)	0.945	0.952	0.970	0.842	34.5
v5	10 Sep	7	104 (74)	0.948	0.955	0.975	0.843	35.1
v6	20 Sep	7	100 (70)	0.941	0.964	0.972	0.844	34.1
best-apron (default)	12 Oct	8	93 (63)	0.971	0.979	0.987	0.864	33.5

The table supports three readings:

- **v2 to v6 differ only in their data.** Every training setting is identical. The time per epoch on the same GPU grew from about 60 s to 94 s, which fits a dataset that grew round by round, as Section 4.2 describes. Each version scored marginally higher in mAP@0.5:0.95 (0.840 to 0.844), and v6 had the best recall (0.964). But each was validated on its own, growing, validation split, so differences of a few thousandths are not evidence that one version beats another.
- **The nano and the apron model were trained on different data and settings.** The nano was trained on Google Colab (NVIDIA T4 GPU) with a shorter schedule, on a separate dataset of recreated working poses only; with no shopfloor images, cloud training stayed within the company's data policy. The apron model used a combined dataset with an eighth class, starting from an earlier fine-tuned run. Neither is directly comparable with the others.

- **None of the six was compared on common ground during training.** Table 7 therefore scores all of them on one test set from station footage, held out from best-apron's training. The scoring is by class name, so differences in class order or class count do not matter. Each image is also judged with the station's own decision logic, to count false permits and false stops directly.

Table 7. All model versions on one common test set (477 images, split `test`). mAP over the 7 classes every model detects; decisions from the station's own logic at its configured floors, single frame. *Like for like* leaves out compliant images that need a class the model lacks (an apron worn instead of sleeves).

Model	Classes	mAP@0.5	mAP@0.5:0.95	Mean recall	False permits	False stops	False stops, like for like	ms/frame
best-apron	8	0.948	0.835	0.951	1.4 % (6/428)	10.4 % (5/48)	10.4 % (5/48)	38.4
v2 (ppe-yolo11s)	7	0.873	0.723	0.796	0.0 % (0/428)	85.4 % (41/48)	72.0 % (18/25)	37.0
yolo11n-nano3	7	0.863	0.711	0.837	0.9 % (4/428)	79.2 % (38/48)	60.0 % (15/25)	22.7
yolo11s-v4	7	0.909	0.770	0.857	0.0 % (0/428)	83.3 % (40/48)	68.0 % (17/25)	37.1
yolo11s-v5	7	0.904	0.772	0.856	0.0 % (0/428)	77.1 % (37/48)	56.0 % (14/25)	36.4
yolo11s-v6	7	0.922	0.808	0.877	0.2 % (1/428)	72.9 % (35/48)	48.0 % (12/25)	36.3

Table 7 settles two questions that the validation scores left open.

The data rounds paid off, and the validation splits hid it. On common ground, mAP@0.5:0.95 rises from 0.723 for v2 through 0.770 (v4) and 0.808 (v6) to 0.835 for best-apron, where Table 6 showed differences of four thousandths. The gains sit in the classes the rounds were aimed at (Tables 8–10): AP@0.5 for gloves rises from 0.77 to 0.97, and recall for safety glasses from 0.55 to 0.90. The eighth class cost the other seven nothing: best-apron leads on five of them and is within 0.011 of the leader on the other two. Safety glasses remain the weakest class for every model.

The decision errors move in opposite directions. best-apron falsely stops 5 of the 48 images labelled compliant (Table 11), against 35 to 41 for the seven-class models. Part of that gap is structural: a model without an apron class must stop every compliant operator who wears an apron instead of sleeves, however well it detects. That applies to 23 of the 48 compliant images. Leaving them out, the seven-class models still falsely stop 12 to 18 of the remaining 25 (48–72 %), against at most 5 for best-apron, so its lead is not an artefact of the apron class.

The like-for-like column also falls steadily from v2 to v6. In the other direction, best-apron falsely permits 6 of 428 non-compliant images, while v2, v4 and v5 permit none. That zero is no merit when it comes with 77–85 % false stops, since a model that rarely permits anything rarely permits wrongly. For a safety system, though, the six false permits matter most, and each must be inspected before best-apron's lead is accepted.

Three limits apply. The counts are per image, from one camera and one frame, while the station combines two cameras over time, so they are not rates per shift. Only 48 test images are labelled compliant, so the 10.4 % false-stop rate has a 95 % Wilson score interval [19] of about 4–22 %, and the 1.4 % false-permit rate one of 0.6–3.0 %. The test split is held out from best-apron, but whether it was also held out from v2–v6, trained on earlier splits of the same images, is not recorded; if not, the older models are flattered, not best-apron.

Table 8. AP@0.5 per class on the common test set. — means the model has no such class.

Class	Instances	best-apron	v2 (ppe-yolo11s)	yolo11n-nano3	yolo11s-v4	yolo11s-v5	yolo11s-v6
person	476	0.987	0.971	0.962	0.985	0.983	0.986
headnet	319	0.994	0.989	0.993	0.995	0.995	0.995
Gloves	421	0.971	0.772	0.814	0.898	0.885	0.904
Mask	281	0.965	0.944	0.916	0.976	0.959	0.970
Safetyglasses	296	0.866	0.724	0.658	0.791	0.769	0.822
sleeves	378	0.923	0.828	0.784	0.842	0.838	0.883
Wrong Sleeve	77	0.928	0.881	0.913	0.873	0.898	0.892
Apron	76	0.990	—	—	—	—	—

Table 9. AP@0.5:0.95 per class on the common test set.

Class	Instances	best-apron	v2 (ppe-yolo11s)	yolo11n-nano3	yolo11s-v4	yolo11s-v5	yolo11s-v6
person	476	0.973	0.891	0.868	0.906	0.928	0.952
headnet	319	0.922	0.862	0.871	0.901	0.867	0.917
Gloves	421	0.866	0.651	0.680	0.758	0.756	0.803
Mask	281	0.849	0.773	0.747	0.830	0.823	0.863
Safetyglasses	296	0.638	0.503	0.454	0.561	0.567	0.605
sleeves	378	0.817	0.695	0.672	0.726	0.729	0.781
Wrong Sleeve	77	0.776	0.684	0.687	0.709	0.735	0.731
Apron	76	0.934	—	—	—	—	—

Table 10. Recall per class at each model's best mean-F1 confidence threshold.

Class	Instances	best-apron	v2 (ppe-yolo11s)	yolo11n-nano3	yolo11s-v4	yolo11s-v5	yolo11s-v6
person	476	0.989	0.916	0.971	0.966	0.964	0.968
headnet	319	0.997	0.950	0.981	0.997	0.988	0.989

Class	Instances	best-apron	v2 (ppe-yolo11s)	yolo11n-nano3	yolo11s-v4	yolo11s-v5	yolo11s-v6
Gloves	421	0.967	0.670	0.753	0.804	0.815	0.862
Mask	281	0.979	0.796	0.897	0.940	0.858	0.900
Safetyglasses	296	0.895	0.554	0.577	0.631	0.675	0.668
sleeves	378	0.934	0.816	0.797	0.828	0.847	0.894
Wrong Sleeve	77	0.896	0.870	0.883	0.831	0.845	0.857
Apron	76	1.000	—	—	—	—	—

Table 11. Station decisions against the labels, per image (single frame, configured floors).

Model	Agree	False permits	False stops	of which need a missing class	Other	Labels: OK / violation / nobody
best-apron	465/477	6	5	0	1	48 / 428 / 1
v2 (ppe-yolo11s)	433/477	0	41	23	3	48 / 428 / 1
yolo11n-nano3	434/477	4	38	23	1	48 / 428 / 1
yolo11s-v4	430/477	0	40	23	7	48 / 428 / 1
yolo11s-v5	435/477	0	37	23	5	48 / 428 / 1
yolo11s-v6	436/477	1	35	23	5	48 / 428 / 1

All five small models cost the same on the CPU (Table 6), so choosing between them is purely a question of accuracy. On that, Table 7 favours `best-apron`, which the station runs.

RF-DETR's accuracy was good, but closing 1500 ms to 40 ms needs about a fortyfold speed-up, and the runtime migration gave about tenfold. The useful conclusion is not that RF-DETR is unsuitable. It is that RF-DETR's accuracy shows **headroom the deployed model does not reach**. The open question is how to capture that headroom inside the latency budget, for example by distillation [20].

5. Inference pipeline

5.1 Threads and stale frames

The pipeline rests on one principle: **no stage waits for a slower one, and no stage works on stale input.** Each camera thread keeps only its newest frame in a one-frame slot, and older frames are overwritten, never queued. The pipeline thread takes only frames it has not processed before. When no camera has delivered for 1.5 s, the station switches to DEGRADED and withdraws the permissive. A queue would have been the conventional design, but the wrong one here. Under load it fills, so every decision is made on an older frame, and for an interlock the age of the frame *is* the response time.

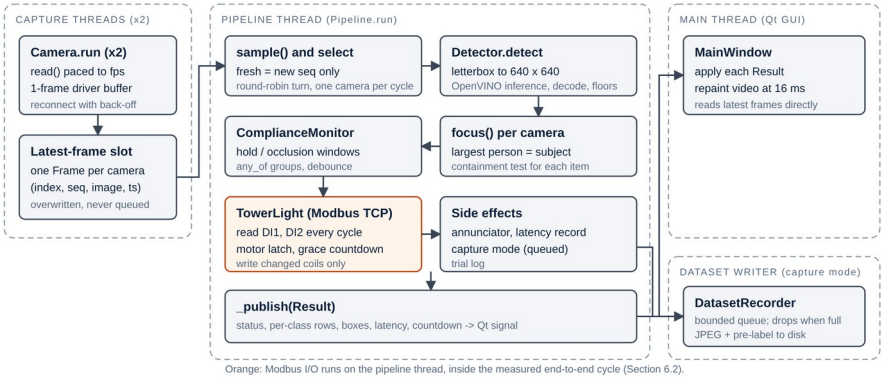


Figure 5. Threads and data flow as implemented.

The Modbus reads and writes run on the pipeline thread (Figure 5). This fixes their order, latch first and lamps second, but puts network round trips inside every cycle. Their cost on the station network has not been measured.

5.2 Camera scheduling

On a CPU, one camera is served per cycle. A batch of two cost 1.89 times a batch of one, and made each frame wait for the other. Taking turns therefore halved the latency per frame: 64 ms against 122 ms on the development host.

A scheduling bug starved the slower camera: the round-robin cursor did not advance when a camera was served alone. A camera with a fresh frame every second cycle was served on only 3 of its 6 chances. After the fix, both cameras are served whenever they have a frame, alternating strictly when both deliver. The resulting scheduler is short:

```
# ppe/pipeline.py:301-306 — camera scheduler (one camera per cycle on a CPU)
if self.detector.batches:
    return fresh
n = len(self.cameras)
chosen = min(fresh, key=lambda f: (f.index - self._turn) % n)
self._turn = (chosen.index + 1) % n
return [chosen]
```

5.3 Letterboxing and USB bandwidth

Frames are *letterboxed* to the square 640 × 640 model input:

- scaled by a single factor;
- centred;

- padded with grey.

Boxes are mapped back by the exact inverse. A direct resize would stretch the image, and several PPE classes are small and are told apart partly by shape.

Two cameras streaming uncompressed video saturated USB. One camera at 1280 × 720 and 30 fps produces about 55 MB/s. Measured on the cameras, 1280 × 720 delivered only 10 fps, while 640 × 480 delivered the full 30 fps. Capture was changed to **640 × 480 at 30 fps**, a threefold improvement in temporal resolution, and **detection accuracy was verified to be unaffected**. That is expected: the letterbox scales a 1280-pixel-wide frame down by half before inference anyway.

Compressed MJPEG transport would have allowed the higher resolution, and I made it work at 30 fps and 720p through the Media Foundation (MSMF) backend. But it needed an operating-system setting on every PC, which would be lost silently whenever IT re-imaged one, so for a system to be replicated across the plant I kept the raw stream. The station captures at 640 × 480 (front) and 640 × 360 (side).

Exporting to OpenVINO cut the time per inference call from 145 ms to 28 ms (5.2×), with detections identical to within 1.2 px. All of this gain came from compilation, because the export stays at FP32. INT8 quantisation [21] remains available, but it needs calibration on site images.

6. Compliance logic and interlock

6.1 From detections to a verdict

Figure 6 shows the decision path from detections to the motor permissive.

The verdict takes one of four values:

- **OK:** the operator is compliant;
- **VIOLATION:** an item is missing, or a forbidden class is present;
- **STANDBY:** nobody is present;
- **DEGRADED:** the station cannot judge.

Only OK permits the motor. In code, the verdict reduces to a few lines:

```
# ppe/tower.py:224-231 — the verdict
def _evaluate(self) -> Status:
    required = [c for c in self.classes if c.required]
```

```
    if not required or self.unavailable():
        return Status.DEGRADED
    if self._subject is not None and not self._subject.present:
        return Status.STANDBY
    return Status.OK if all(c.satisfied for c in required) else
Status.VIOLATION
```

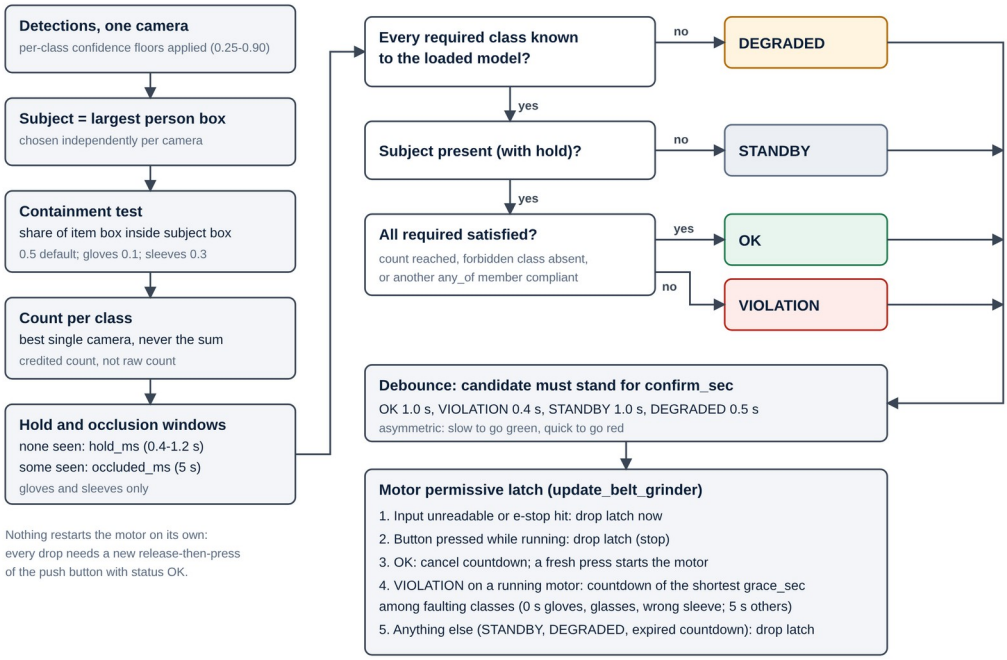


Figure 6. Decision flow from detections to the motor permissive.

Two choices recur throughout. First, **counts come from the best single camera, never the sum**: two cameras that each see the same two gloves must not report four. Second, **the operator is the largest person in each camera's view**, which is a cheap and robust proxy for being closest to the machine. An item counts only if a large enough share of its box lies inside the operator's box:

- 50 % in general;
- 30 % for sleeves;
- 10 % for gloves, which sit at the ends of outstretched arms beyond the person box.

Bystanders and their equipment are ignored. A missed person gives STANDBY, which darkens the lamp but still stops the motor. The same mechanism explains the large-workpiece problem in Section 8.3.

Figure 7 shows both rules on photographs of a recreated working pose, taken in an isolated environment and annotated with best-apron's detections at the station's confidence floors. The

side view (a) holds every required item and on its own would permit the motor; only one sleeve is visible, which the station's count of one accepts. The front view (b) shows the head net, glasses and mask clearly but not the hands, so on its own it would be a violation. Judged together, as the station judges its two cameras, the pair is compliant: the gloves and sleeve come from the side camera.



Figure 7. Detections by best-apron on a recreated working pose: (a) side view, (b) front view, each a single frame at the station's confidence floors.

6.2 The debounce and hold-window trade-off

Early tests showed the person box, and the detections that depend on it, fluctuating from frame to frame. Driven directly, the interlock would have chattered against a workpiece held to the belt. Two mechanisms stabilise the verdict, and they pull in opposite directions.

The debounce applies a new status only after it has stood for a confirmation period. After senior management raised response time at the first demonstration, the debounce was *reduced*. The original counted three agreeing cycles. That was about 1 s at PyTorch cycle rates, but varied with CPU load. It was replaced by periods in seconds that differ by status:

- 0.4 s to go red;
- 1.0 s to go green;
- 1.0 s to conclude that nobody is there.

The station is quick to alarm and slow to claim compliance. The cost is stability: a shorter debounce lets briefer detection gaps through.

Per-class hold windows restore the stability, but only where it is needed. A class that drops out keeps its credited count for its hold period, from 0.4 s to 1.2 s. **Gloves and sleeves** also get a separate 5 s *occlusion window* that applies only while *some but not all* of the pair are visible. One visible glove is evidence that the pair is worn, so the tolerance is granted on evidence. Safety glasses, rarely occluded the same way, get only their 1.1 s hold. Tests confirmed that the revised logic handles occlusion of gloves and sleeves well. Figure 8 shows both windows for a pair of gloves.

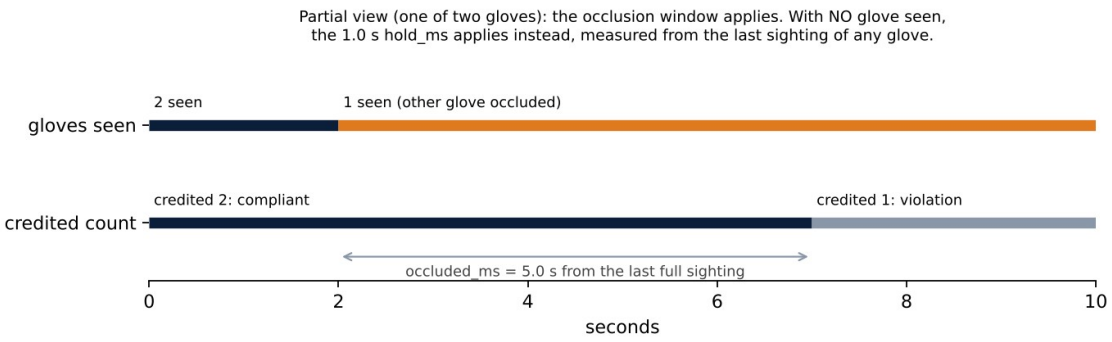


Figure 8. The two windows for a pair of gloves.

Testing also found a defect, which I measured rather than assumed. Both windows had been timed from the last *full* sighting, so a partial view silently consumed the dropout hold. After one second of seeing a single glove, the configured 1.0 s hold was zero, on exactly the classes the occlusion window was added for. Timing each window from its own evidence fixed it.

The trade-off has a cost. An operator who removed one glove while the other stayed visible was still credited for up to about 5.4 s. This bounded exposure was accepted in exchange for not stopping operators whose second glove was behind the workpiece.

Station tuning has since moved this trade-off further. The station configuration now requires **one** glove and **one** sleeve, not two. That removes the false stops of the side view, which can rarely see both hands at once. Because a set of one cannot be partly visible, the occlusion window, now 8 s, no longer applies. The cost is that **an operator with one bare hand is judged compliant indefinitely**: in simulation, removing one of two gloves never stopped the motor. The station also raised:

- the person confidence floor from 0.25 to 0.70, which suppresses background people at the cost of losing the operator more easily;
- the `Wrong Sleeve` hold from 0.4 s to 1.2 s.

These are risk decisions, not only tuning decisions, and they belong in the risk assessment. Table 12 lists every class's thresholds and timings.

Table 12. Per-class settings of the station configuration, and the time from an item leaving view to the motor stopping, simulated with the station's own logic on a running machine. Values from the first configuration are given in brackets where they changed.

Class	Confidence	Containment	Hold	Count	Correction window	Motor stops after
Head net	0.50	0.5	0.8 s	1	8 s (5 s)	9.3 s (6.3 s)
Safety	0.60	0.5	1.1 s	1	5 s (0 s)	6.6 s (1.6 s)

Class	Confidence	Containment	Hold	Count	Correction window	Motor stops after
glasses						
Mask	0.60	0.5	0.7 s	1	8 s (5 s)	9.2 s (6.2 s)
Gloves	0.67 (0.60)	0.1	1.0 s	1 (2)	5 s (0 s)	6.5 s (1.5 s); one of two removed: never (5.5 s)
Sleeves	0.75	0.3	1.2 s	1 (2)	8 s (5 s)	9.7 s (6.7 s); one of two removed: never (10.5 s)
Apron	0.60	0.5	1.2 s	1	8 s	as sleeves (alternative)
Wrong Sleeve (forbidden)	0.90	0.5	1.2 s (0.4 s)	0	0 s	0.5 s after it appears
Person (subject)	0.70 (0.25)	—	1.0 s	—	—	STANDBY, motor stops after ≈ 2 s

Other settings: confirmation periods OK 1.0 s, VIOLATION 0.4 s, STANDBY 1.0 s, DEGRADED 0.5 s; occlusion window 8 s (5 s), which no longer applies with a count of one; runtime NMS IoU 0.50; global confidence floor 0.35.

6.3 Warning before inhibition

A violation on a *running* machine opens a **correction window** instead of stopping it at once. The lamp turns red, the buzzer warns, and the screen counts down. If the operator puts the item back, the machine never stops. Otherwise the permissive is withdrawn at the end of the window.

This is a **deliberate safety trade-off: for the length of the window the machine runs with a detected violation.** As implemented, the window has four properties:

- it was first set at 5 s, with no window at all for safety glasses, gloves and Wrong Sleeve, whose absence *is* the immediate hazard;
- on the station it is now **8 s by default, 5 s for safety glasses and gloves**, and still **0 s for Wrong Sleeve**;
- the shortest window among the current violations applies, so a missing head net cannot shelter a missing glove;
- it is set once and never renewed, so flickering PPE cannot hold the machine open.

Figure 9 shows the resulting stop times. I obtained them by running the station’s compliance and latch code on its configuration with simulated detections. The same method reproduces

the times recorded earlier for the first configuration (1.5 s for gloves, 6.3 s for a mask). A detected wrong sleeve still stops a running machine in 0.5 s. But a missing glove or missing safety glasses now keep it running for about 6.5 s, and a missing mask, head net or sleeves for about 9.5 s. The station is configured this way to let an operator correct a glove that has slipped. That is a defensible choice for nuisance stops, but it reverses the earlier decision to give no window to the items whose absence is the immediate hazard. **It needs an explicit, documented EHS position before rollout.**

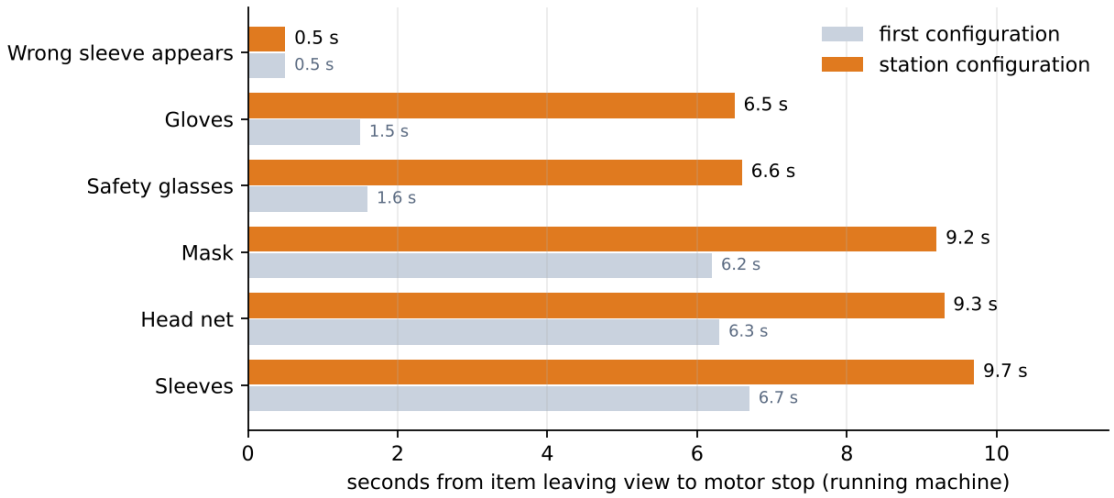


Figure 9. Time from an item leaving view to motor stop, station configuration (simulated with the station's own logic).

The permissive is a **latch**, like the seal-in contact of a motor starter. Pressing the button starts the motor only when four conditions hold:

- the operator is compliant;
- the e-stop is not pressed;
- the press is a genuine release-then-press edge that the software observed;
- the machine is idle.

Pressing the button while the motor runs stops it. Nothing restarts the motor on its own. The coils are written only when they change, the inputs are polled every cycle, and every output is switched off at connection and at shutdown.

6.4 Failure behaviour

I analysed what the code does when each part fails (Table 13), and confirmed two paths by running the real pipeline against a stubbed model. Both were software faults, and I fixed them. The behaviour divides as follows.

Fail-safe paths:

- loss of all cameras;
- a model that fails to load;
- an unreadable input;
- the operator leaving the cell.

Paths that could leave the grinder enabled while detection was not working:

1. **One camera stops while the other runs (fixed).** Its last detections stayed in force indefinitely: in simulation the status stayed OK while the live camera saw no gloves. Now a camera without a new frame for 1.5 s counts as down, its results are discarded, and the station goes DEGRADED, naming the camera on screen. A camera that keeps streaming a *frozen image* still looks live to the software.
2. **An exception during inference (fixed).** The pipeline thread ended without switching the relay off, and the window stayed open, so the supervisor did not restart it. Now every failed cycle switches the motor off; after five in a row the station closes the relay connection and exits with an error code, so the supervisor restarts it.

Each fix came with a test that fails on the old code. The remaining paths need hardware or device configuration rather than code:

3. **Loss of communication, a hang, or a PC crash or power loss.** The relay module keeps its last outputs unless its own communication watchdog is enabled. The deployment guide relies on that watchdog, but nothing records that it was tested.
4. **The emergency stop depends on the software path.** Whenever that path fails or stalls, the e-stop cannot stop the motor either, and a broken wire to its normally open contact goes unnoticed. ISO 13850 [22], IEC 60204-1 [23] and IEC 60947-5-5 [24] require a hard-wired e-stop with normally closed, direct-opening contacts, and the safety-relay circuit should be designed and validated to a performance level determined under ISO 13849-1 [25]. **Until the safety relay is fitted, the station's e-stop should not be relied upon as an emergency stop.**

None of these concerns the model. Each lies at the boundary between software and machine, where a permissive must be most conservative.

Table 13. Behaviour of the implemented code under failure.

Failure	What the code does	Grinder ends up
Both cameras stop	DEGRADED after 1.5 s stale + 0.5 s confirm; latch dropped	Inhibited (≈ 2 s)
One camera stops, other	Camera counted down after 1.5 s; its results	Inhibited (≈ 2 s; fixed)

Failure	What the code does	Grinder ends up
runs	discarded; DEGRADED, camera named	
Camera streams a frozen image	Treated as fresh frames	**Can stay enabled**
Model fails to load	Board blanked before load; pipeline closes the board	Inhibited
Inference throws mid-run	Each failed cycle sets DEGRADED; after 5 in a row, board closed and process exits	Inhibited; supervisor restarts (fixed)
Relay unreachable	Latch dropped in software; write cannot arrive	**Last state** unless module watchdog
Crash or kill	Supervisor restarts after ≥ 2 s; board blanked on connect	**Enabled until restart**
Hang, PC power loss, cable pulled	Nothing detects it	**Last state** unless module watchdog
E-stop pressed	Read on DI1; latch dropped	Inhibited, via software only
E-stop wire broken (normally open)	Reads as “not pressed”	**E-stop ineffective**
Operator leaves	STANDBY after ≈ 2 s	Inhibited

7. Deployment and commissioning

7.1 Hardware and commissioning

I researched industrial PCs against two sets of requirements:

- **compute:** a multi-core x86 CPU without a discrete GPU, as the measurements had shown;
- **the shopfloor:** dust, panel mounting, and two separate USB ports for the cameras.

I obtained quotations, and evaluated the IT department’s standard configuration so that the station would sit inside existing support arrangements. The chosen unit was tested on the shopfloor before commissioning: a compact PC from the IT standard range (Intel Core i5-13500T, 14 cores, 16 GB RAM, Windows 11 Enterprise), whose processor cores run the inference.

Electrical commissioning was done with the maintenance department. I verified the wiring of the vendor-built panel (tower light, relay module, e-stop and push button) before installation. Modbus was commissioned in **two separate stages**:

1. **Network connectivity.** The module is reachable and its Modbus TCP port accepts connections.
2. **Protocol-level communication.** Using `pymodbus`, I wrote each coil, confirmed the physical actuation, and read the inputs with the switches operated.

Separating the stages makes failures unambiguous. A module can accept a TCP connection and still drive the wrong relay, because Modbus addresses start at 0 while the module's labels start at 1.

7.2 Installation and relocation

The station was installed at one grinder. Three items of scope remained at that point:

- camera cable routing;
- the overload relay;
- mounting the PC in the panel.

Management then decided to move the installation to a more accessible position. That required a new Management of Change request, measurement of the new location, and re-finalisation of the camera positions from the model's performance there. Maintenance carried out the move.

The relocation **exposed a dependency on camera geometry**: in the side view, the model began to miss gloves, glasses and masks. Capturing training images from varied positions limited the damage, but did not eliminate it.

7.3 Safety approval, documentation and reviews

The system went through the site's **Management of Change** (MOC) process. An EHS meeting reviewed its features, the station's existing safety practices and the expected requirements, and identified scope for further development. A **second MOC** covered the relocation. I obtained the station's **EHS Risk Assessment** from the responsible High Performance Team, updated it to reflect the system, and submitted it for approval. I wrote a **Standard Operating Procedure** for operator training, which was reviewed and revised. A 3D model of the installation was requested from the design department.

The system was demonstrated to four audiences in turn. Each review changed something.

1. **Shopfloor staff.** Their feedback shaped later iterations.
2. **Site senior management.** They raised the size of the control panel and the response time.
3. **Site executives.**
4. **The CEO of Unison Industries,** a GE Aerospace subsidiary.

The plant then decided to deploy the system on all flat belt grinders, and to present it as a **safety moment to the global CEO of GE Aerospace**.

8. Results and evaluation

8.1 Latency

End-to-end latency on the station fell from 300–400 ms to about 40 ms, measured from camera grab to relay write. A controlled benchmark on a 4-core development host reproduced the effect (Figure 10):

- the median fell from 488 ms to 48 ms;
- the cycles completed in 25 s rose from 56 to 793.

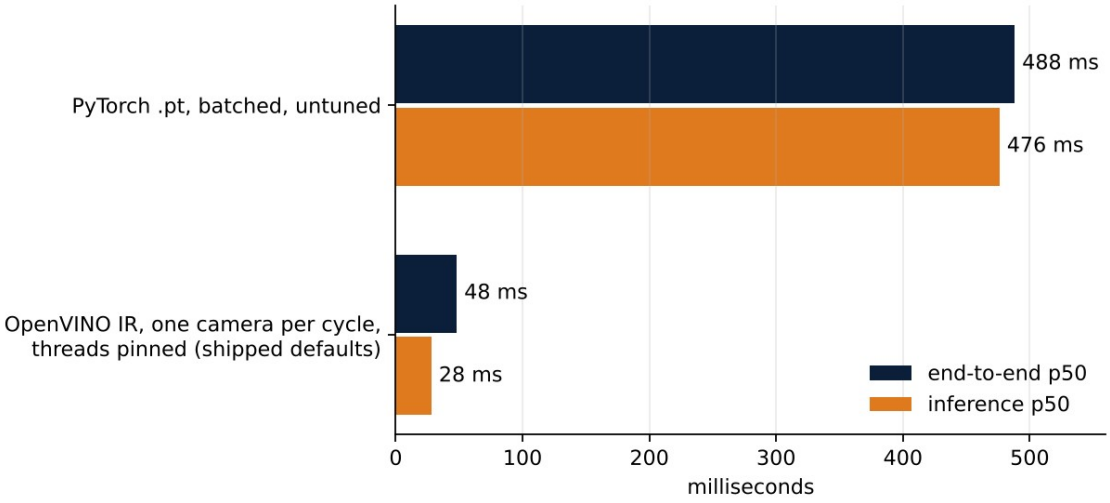


Figure 10. Median end-to-end and inference latency before and after optimisation (development host).

Per-stage instrumentation decided the fix. In the untuned pipeline, inference took **98.6 %** of each cycle, and all the code I had written myself took under 2 %. Only the forward pass was worth optimising, through the runtime, the model size or the input size. The second-largest gain came from noticing that two measurements which should have agreed did not: inference inside the application was three times slower than in the benchmark, because the capture threads' OpenCV pools competed with the model for the cores. Pinning capture to one thread per camera closed the gap. One qualification applies. The pipeline grew while it was being optimised, so the intermediate measurements are not comparable, and I report only the two endpoints measured under identical conditions.

8.2 Detection performance

The validation figures (P 0.942, R 0.955, mAP@0.5 0.966) are strong for a seven-class site detector, but they carry four limits: they are averages over the classes, they come from the split used to select the model, they share their viewpoints with the training data, and they measure detection, not decision.

The per-class results remove the first limit (Tables 8–10, and best-apron's validation below), and the common test set the second: there, v2 scores an mAP@0.5 of 0.873 rather than 0.966, and best-apron 0.948 (Table 7). Neither set of figures says anything about carried gloves or prescription spectacles. In operation the model is no longer the main source of risk. The false stops that mattered came from viewpoints it was not designed for, and the false permits come from decisions the schema cannot express (Section 4.2) and from the software–machine boundary (Section 6.4).

Validation of best-apron. Figures 11 to 13 show the final validation of the eight-class model the station runs. Per-class AP@0.5 ranges from 0.972 for safety glasses to 0.995 for the head net and mask, with a mean of 0.986 (Table 6 records 0.987 at the best epoch during training). The F1 score over all classes peaks at 0.97 at a confidence of 0.46 and stays within a few hundredths of that value between about 0.1 and 0.7, so the result is not sensitive to the exact threshold. Safety glasses are the exception: their F1 score is the lowest at every threshold and the first to fall away, and their precision drops below about 0.9 at confidences under 0.3, as does that of `Wrong Sleeve` to a lesser degree. The station's floors for these two classes, 0.60 and 0.90 (Table 12), lie well above that region.

The confusion matrix shows almost no confusion between classes: at least 97 % of the instances of each class are detected as that class. The errors lie with the background. Up to 3 % of gloves and wrong sleeves are missed, and of the false detections on background, 35 % are safety glasses and a quarter each are gloves and sleeves. A false detection of a required item is the error that can credit missing PPE; the containment test of Section 6.1 discards those outside the operator. The only confusion between two classes is that 1 % of wrong sleeves are detected as sleeves. It is small but safety-relevant, because a wrong sleeve read as a sleeve is a path to a false permit. The training curves (Figure 13) show the validation losses levelling off after about 60 epochs, consistent with the best epoch of 63.

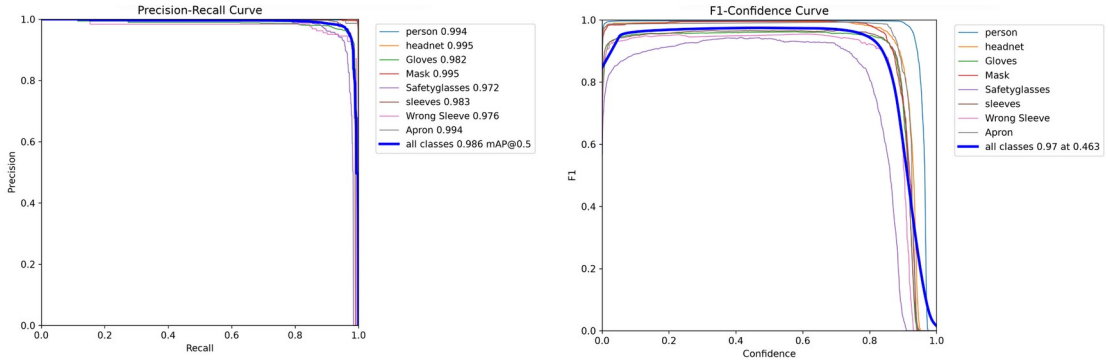


Figure 11. Precision–recall and F1–confidence curves of best-apron. Validation split.

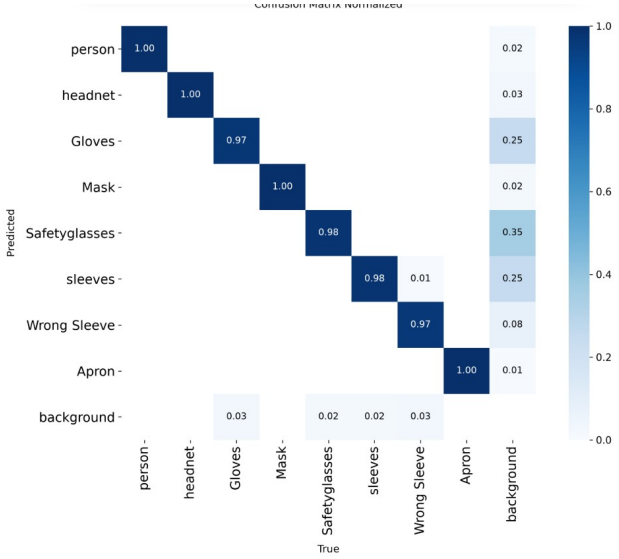


Figure 12. Normalised confusion matrix of best-apron. Validation split; each column is a true class.

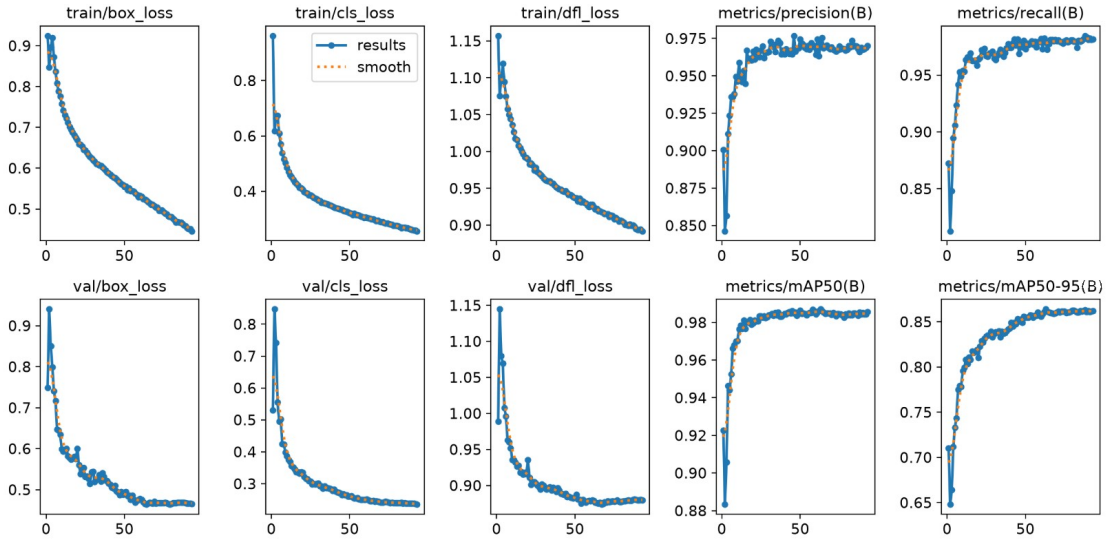


Figure 13. Training curves of best-apron over 93 epochs: losses and validation metrics.

8.3 The operator feedback finding

The most consequential result came from watching the deployed system in use and talking to the operators. Three things emerged.

- **Posture.** Operators customarily stand *at an angle* to the machine, to inspect the flatness of the trim. That is outside the cameras' favourable viewing angle, so items are missed and the machine stops falsely.
- **Aprons.** Operators want to wear aprons against grinding dust, and the model had no apron class.
- **Large workpieces.** Larger parts force the operator to stand further back, partly out of frame. The station then finds nobody, and the machine cannot start.

Three remedies were identified:

- reposition the cameras for large workpieces;
- angle them so that every class stays observable in the posture operators actually adopt;
- retrain with an apron class. This has since been done: the station runs `best-apron`, an eight-class version (Table 6), and its configuration accepts an apron in place of sleeves.

This is a requirements failure, not a model failure. The model correctly reported that it could not see a glove turned away from it. The camera placement had been derived from an *assumed* posture, not the *observed* one. Every downstream step inherited that assumption, including the validation set, so no metric could reveal it. The finding generalises: across the plant, **camera placement cannot be specified from the machine layout alone**. It must be derived from how each operation is actually performed, which is why the rollout began with a gemba walk.

8.4 Evaluation against the objective

Section 1.3 set four requirements. Measured against them, the system meets two in full, one in part, and one only with qualifications that the rollout must close.

- **It withholds the run permissive.** Met. Only the OK status permits the motor. Every other status, an unreadable input and the e-stop all withdraw it. Nothing restarts the motor without a fresh press of the button. The correction window delays the stop on purpose for every item except a wrong sleeve, by 5–8 s. That delay is a documented trade-off, but one that still needs its EHS position.
- **It judges the operator at the machine.** Met, with one known limitation. The largest-person filter and the containment test ignore bystanders and their equipment, but the person *nearest the camera* is judged. That is usually, though not necessarily, the person at the belt.

- **It responds quickly enough.** Met. At about 40 ms, the pipeline is no longer what limits the response. The station declares a violation in 0.4 s plus the class's hold. Its response time is now a choice made in the logic rather than a limit set by the hardware, which answers management's concern directly.
- **It is steady enough to be accepted.** Partly met. The hold and occlusion windows, and later the single-glove requirement, removed the chatter caused by occluded gloves and sleeves. The false stops caused by the angled working posture and by large workpieces remain, until the cameras are repositioned.

Two properties the objective took for granted are not yet met:

- **Fail-safe behaviour at the boundary between software and machine** (Section 6.4). The two software faults are fixed; the hard-wired e-stop and the relay watchdog are not.
- **Detection of decision failures** such as carried gloves (Section 4.2).

Neither is a matter of model accuracy, and both are taken up as further work in Chapter 11.

9. Plant-wide rollout

The rollout to every flat belt grinder at the plant is to be carried out through an external supplier. I developed the specification package over several rounds of review:

- **The RFQ.** I added a *sequence of operations* to it, because the system is not a catalogue item and the split of responsibility between supplier and site had to be unambiguous.
- **A Bill of Quantities and Scope of Work** for each station.
- **Acceptance test protocols**, with a pulled-wire test on each switch input.

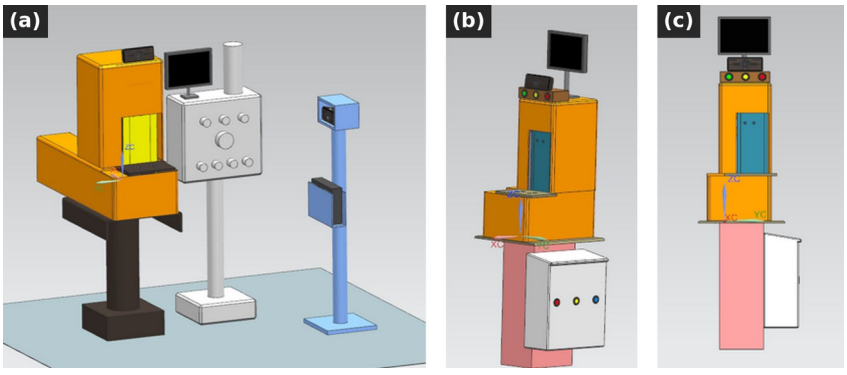


Figure 14. Layout models of the station. (a) Prototype, with separate stands for the control panel and monitor and for a camera. (b, c) Integrated version: monitor and tower light above the grinder, control panel on its pedestal.

I also evaluated industrial cameras to replace the consumer webcams, which are not defensible in a dusty environment. The specification asked for USB 3 direct to the PC on a 2 m cable, and working distances of 0.5–1.0 m for the front camera and 1.0–1.5 m for the side camera. A **gemba walk** surveyed every grinder in the condition in which it actually operates.

Standardisation rests on a single self-testing Windows installer whose only per-station input is the machine number. Here two earlier decisions prove their value: rejecting the machine-dependent MJPEG setting saves an operating-system change at every station, and local annotation lets every station contribute failure cases without an exception to the data rules.

The station was also redesigned as an integrated tower that fits within the grinder's own envelope (Figure 14): 0.76 m³ against 1.63 m³ for the prototype's separate units, a 53 % reduction in volume. The monitor and tower light move onto the grinder, and the control panel onto its pedestal, so nothing stands on the floor beside the machine. This answers management's comment on the size of the panel.

10. Critical reflection

10.1 Supervision

I worked closely with my industrial supervisors. The work was divided into smaller tasks, each reviewed before the next phase began. Because the system acts on a machine on the shopfloor and is safety-critical, this close supervision was both appropriate and necessary. I also worked within a cross-functional team with members from EHS, maintenance, IT and lean management, whose contributions shaped the project throughout and were essential to its success.

10.2 Learning achievements

Deployment is most of the work. The model passed 0.96 mAP@0.5 early. Most of the internship was spent on what lies between that number and an operator starting a machine.

Diagnose before optimising. Instrumentation showed inference at 98.6 % of the cycle. Without it I would have optimised my own code, which cost under 2 %.

Requirements come from watching the operation. No metric revealed the angled working posture. Observation and conversation did, just as observation had produced the project in the first place.

Take the deployment constraint seriously, even when it costs performance. The MSMF route worked, and rejecting it still proved the right decision once the rollout was in view.

A safety system lives inside a process. The system became a control when the risk assessment described it, not when the code ran. The questions EHS asked (what happens when a camera, the network or the PC fails?) led to the analysis that found my own code's weakest points.

Different audiences need different accounts of the same system. Explaining a part of it to someone who would never read its code was the best test of whether I understood it.

10.3 Problems and what I would do differently

- **Camera placement was decided too early.** I would observe operators at work before fixing the camera positions and before capturing training data, not after deployment.
- **Requirements drifted away from the documents.** The correction window, the switch wiring and the scheduler all changed after they had been documented, and the specifications had to follow. Keeping one owner for those documents, and a record of each change, would have saved review rounds.
- **Safety integration lagged behind the software.** The hard-wired e-stop remains further work, and two software failure paths were found only in the final review. A failure-mode analysis written *before* the first installation, not after it, would have found them earlier.

11. Conclusion and further work

The internship delivered a working, approved and replicable PPE interlock:

- a two-camera, CPU-only station with about 40 ms latency;
- a site-specific detector, now with eight classes, compared with its predecessors on one test set;
- compliance logic that separates detection from decision and balances responsiveness against stability class by class;
- two MOC approvals, an updated risk assessment and an SOP;
- two fail-safe faults found in review and fixed, each with a test;
- a rollout package for every flat belt grinder at the plant.

Further work would build on these results; Table 14 lists it in order of priority.

Table 14. Further work, in order of priority.

#	Area	Proposed work
1	Hard-wired emergency stop	Fit the safety relay, with normally closed, direct-opening contacts in the contactor circuit
2	Communication watchdog	Enable the relay module's watchdog and test it by disconnecting the network cable with the motor running
3	Correction windows and counts	Document the EHS position on 5 s for gloves and glasses, 8 s for the other items, and one glove instead of two
4	Apron model	Examine its six false permits on the common test set (Table 7); confirm in the risk assessment that an apron may replace sleeves
5	Camera geometry	Reposition and angle the cameras for the observed working posture
6	Frozen camera image	Detect a camera that keeps streaming an unchanging image, for example by checking that consecutive frames differ
7	Spectacles and carried gloves	Build a targeted evaluation set; add a relational test of each glove against the detected hands
8	Modbus timing	Measure the relay stage's contribution to latency on the station
9	Default capture configuration	Align the shipped default with the station (640 × 480) before rollout
10	Library versions	Align the training, export and runtime versions at the next retraining
11	Test-split independence	Confirm that the common test images were also held out from v2–v6
12	Integrated graphics	Benchmark OpenVINO on the processor's integrated GPU, which the station does not yet use

In the longer term, the latency headroom could be used to capture the accuracy that RF-DETR showed is available; a longer shop-floor trial that logs false stops and false permits would measure what the per-image evaluation can only estimate; decision failures could be made measurable with a targeted evaluation set and then addressed with a relational test of each glove against the detected hands; the capture mode could feed periodic retraining; stop records aggregated across the fleet could inform tuning; and the approach could be applied to other manual operations, starting each time from how the operation is actually performed.

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